GTU ROBOTICS CLUB

Docker setup:

1.Start the basic knowledge of docker
2.Setup the docker in your system :
https://youtu.be/cqbh-RneBlk?si=WGxigQtl1LdN9a9W

3.Set the Nvidia - smi in your host system for gpu access
See the docs of Tensorrt_cuda_deployment_pdf in GITHUB

4.Set the nvidia-container toolkit
https://docs.ultralytics.com/guides/docker-quickstart/#prerequisites
Go through from this website
Verify NVIDIA-SMI Command
Start the setup of NVIDIA container toolkit from this website

5. Find the best docker page
LINK: https://hub.docker.com/r/ultralytics/ultralytics/tags
Example : Pull this image
Cmd: docker pull ultralytics/ultralytics:latest-python-export

6. Check through command
Cmd: **sudo docker images**(for images)
Cmd: **sudo docker ps -a**(for container)

7.enter the docker images with gpu , usb, photo, video,

Start with photo
Cmd : echo $DISPLAY
Cmd: # On host
xhost +local:root (for photo, gui access everytime run with or enter the container after above two cmd run)

8. Enter the setup of NVIDIA-SMI host system as well as then into the docker
A.cuda
B.Nvidia-smi
C.cudnn
D.Tensorrt

File from host to into the docker:
Cmd: sudo docker cp /home/purushottam/Downloads/nv-tensorrt-local-repo-ubuntu2204-10.13.2-cuda-13.0_1.0-1_amd64.deb ultralytics_container:/ultralytics/

9. Start this basic configuration
apt-get update
apt-get install -y usbutils
Check cmd: lsusb

Start the downloading and installing library
opencv -python, pyrealsense and other necessary libraries

Install the nano for edit and make a new file
apt-get update
apt-get install -y nano

Basic nano shortcuts

- **Ctrl + O** → Save the file (writes out)
- **Enter** → Confirm filename when saving
- **Ctrl + X** → Exit nano
- **Ctrl + K** → Cut a line
- **Ctrl + U** → Paste a line
- **Ctrl + W** → Search text

Go through this website for docker as well as host system for Realsense SDK

Link :

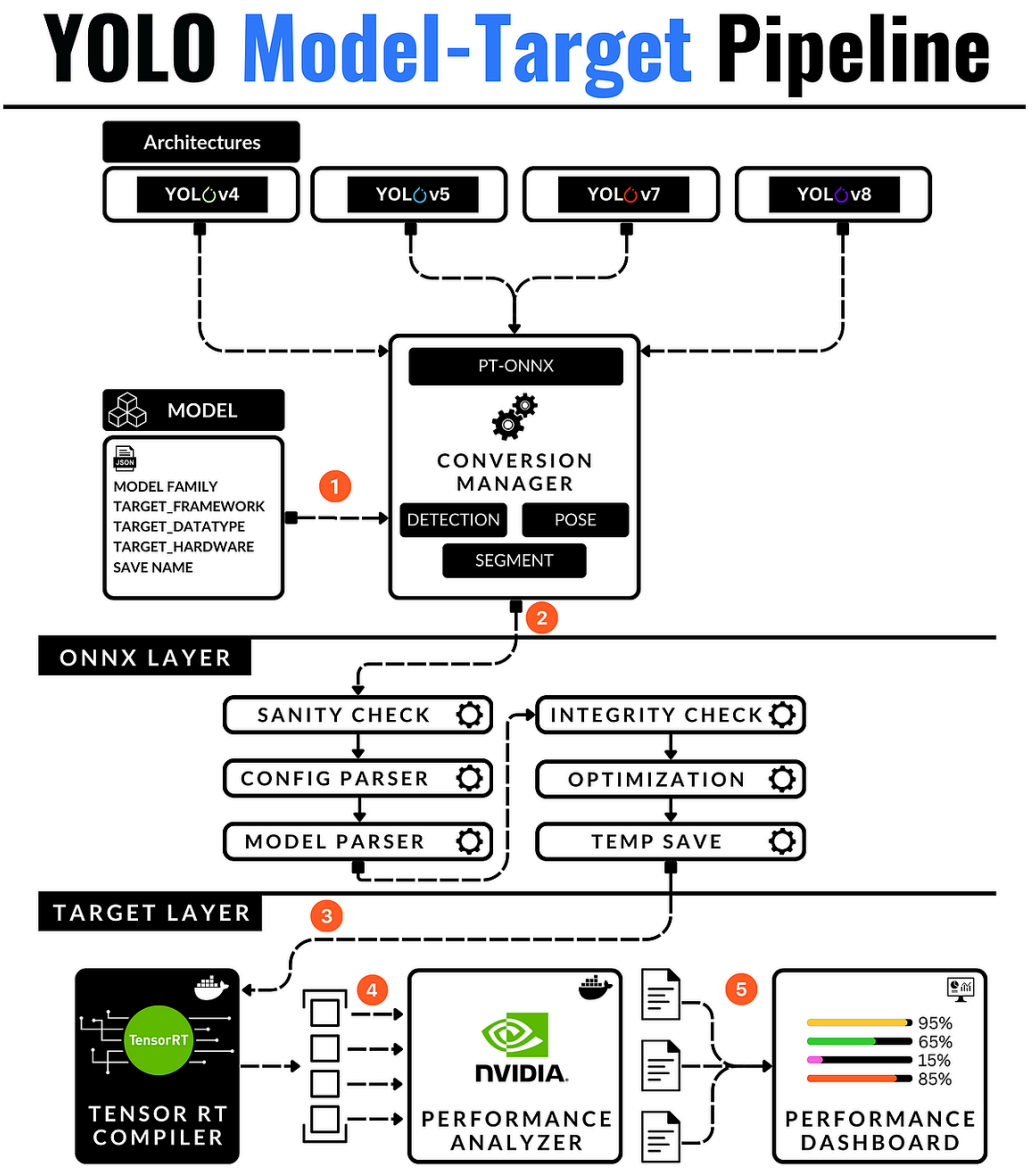
https://github.com/IntelRealSense/librealsense/blob/master/doc/distribution_linux.md

```
root@0694fdbd00f0:/ultralytics# lsusb
Bus 002 Device 011: ID 8086:0b5c Intel(R) RealSense(TM) Depth Camera 455  Intel(R) RealSense(TM) De
Bus 002 Device 001: ID 1d6b:0003 Linux 6.8.0-83-generic xhci-hcd xHCI Host Controller
Bus 001 Device 003: ID 048d:c966 ITE Tech. Inc. ITE Device(8176)
Bus 001 Device 002: ID 5986:212b SunplusIT Inc Integrated Camera
Bus 001 Device 004: ID 8087:0026
Bus 001 Device 001: ID 1d6b:0002 Linux 6.8.0-83-generic xhci-hcd xHCI Host Controller
root@0694fdbd00f0:/ultralytics# ls /dev/video*
/dev/video0  /dev/video1  /dev/video2  /dev/video3  /dev/video4  /dev/video5  /dev/video6  /dev/vid
root@0694fdbd00f0:/ultralytics# ls
2025-09-23-161706.webm  README.zh-CN.md                                                docker
CITATION.cff            ai.jpeg                                                        docs
CONTRIBUTING.md         best.pt                                                        examples
LICENSE                 cuda-repo-ubuntu2204-13-0-local_13.0.1-580.82.07-1_amd64.deb  mkdocs.yml
README.md               cudnn-local-repo-ubuntu2204-9.13.0_1.0-1_amd64.deb             nv-tensorrt-l
root@0694fdbd00f0:/ultralytics# python3 realsense_test.py
root@0694fdbd00f0:/ultralytics# nano trained_model.py
root@0694fdbd00f0:/ultralytics# python3 trained_model.py

0: 480x800 (no detections), 49.7ms
Speed: 3.7ms preprocess, 49.7ms inference, 16.0ms postprocess per image at shape (1, 3, 480, 800)
```

Check above all command working in inside docker container and run your first code with ultimate GPU access.(plug intel realsense as well)

Most Imp How to set TensorRT Framework in Docker
Use this code to convert it .pt file to .engine file
Cmd: python3 convert.py

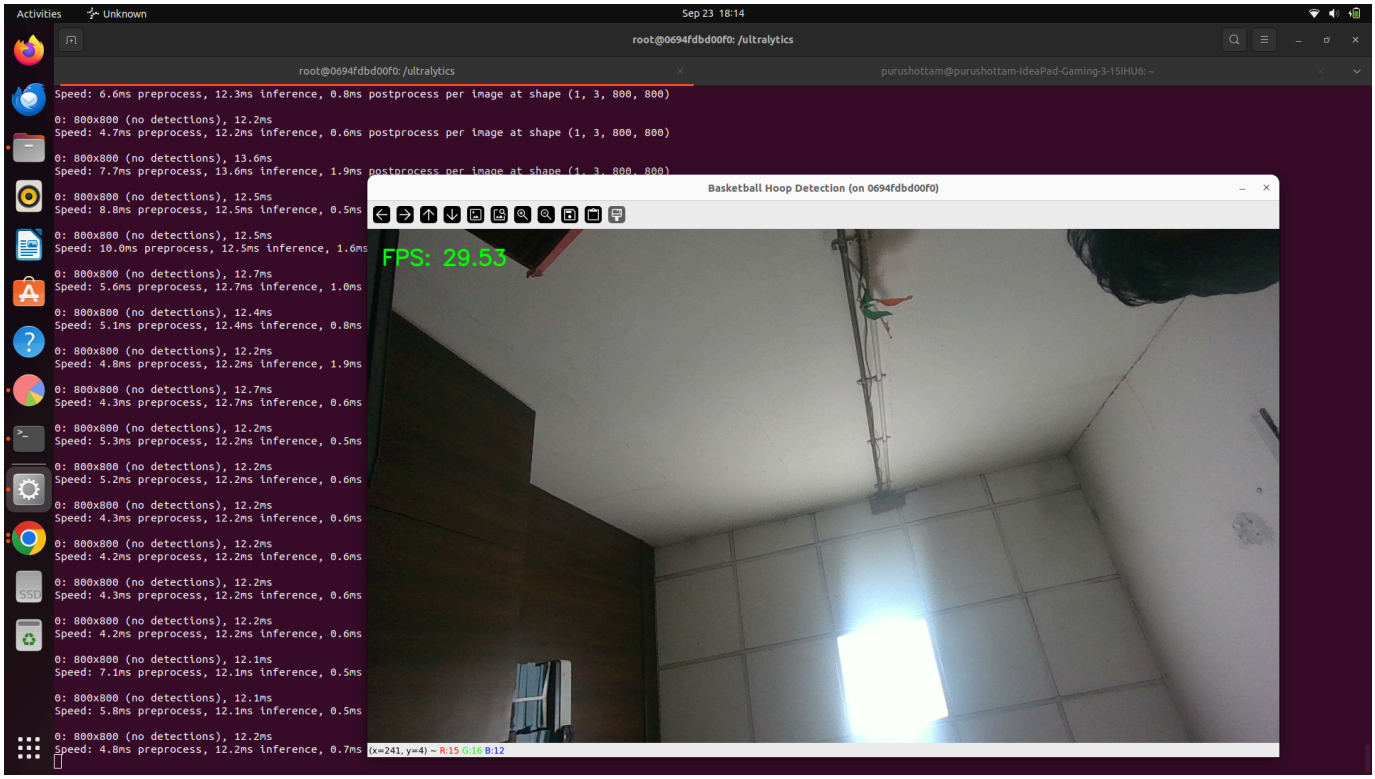
```python
from ultralytics import YOLO

# Load a YOLO11n PyTorch model
model = YOLO("yolo11n.pt")

# Export the model to TensorRT
model.export(format="engine")  # creates 'yolo11n.engine'

# Load the exported TensorRT model
trt_model = YOLO("yolo11n.engine")

# Run inference
results = trt_model("https://ultralytics.com/images/bus.jpg")
```



After that run you file

Cmd: python3 trained_model_with_TensorRt.py

You see something like this


Activities
Unknown
Sep 23 18:14
root@0694fdbd00f0: /ultralytics
root@0694fdbd00f0: /ultralytics
purushottam@purushottam-IdeaPad-Gaming-3-15IHU6: ~
Speed: 6.6ms preprocess, 12.3ms inference, 0.8ms postprocess per image at shape (1, 3, 800, 800)
0: 800x800 (no detections), 12.2ms
Speed: 4.7ms preprocess, 12.2ms inference, 0.6ms postprocess per image at shape (1, 3, 800, 800)
0: 800x800 (no detections), 13.6ms
Speed: 7.7ms preprocess, 13.6ms inference, 1.9ms postprocess per image at shape (1, 3, 800, 800)
0: 800x800 (no detections), 12.5ms
Speed: 8.8ms preprocess, 12.5ms inference, 0.5ms
0: 800x800 (no detections), 12.5ms
Speed: 10.0ms preprocess, 12.5ms inference, 1.6ms
0: 800x800 (no detections), 12.7ms
Speed: 5.6ms preprocess, 12.7ms inference, 1.0ms
0: 800x800 (no detections), 12.4ms
Speed: 5.1ms preprocess, 12.4ms inference, 0.8ms
0: 800x800 (no detections), 12.2ms
Speed: 4.8ms preprocess, 12.2ms inference, 1.9ms
0: 800x800 (no detections), 12.7ms
Speed: 4.3ms preprocess, 12.7ms inference, 0.6ms
0: 800x800 (no detections), 12.2ms
Speed: 5.3ms preprocess, 12.2ms inference, 0.5ms
0: 800x800 (no detections), 12.2ms
Speed: 5.2ms preprocess, 12.2ms inference, 0.6ms
0: 800x800 (no detections), 12.2ms
Speed: 4.3ms preprocess, 12.2ms inference, 0.6ms
0: 800x800 (no detections), 12.2ms
Speed: 4.2ms preprocess, 12.2ms inference, 0.6ms
0: 800x800 (no detections), 12.2ms
Speed: 4.3ms preprocess, 12.2ms inference, 0.6ms
0: 800x800 (no detections), 12.2ms
Speed: 4.2ms preprocess, 12.2ms inference, 0.6ms
0: 800x800 (no detections), 12.1ms
Speed: 7.1ms preprocess, 12.1ms inference, 0.5ms
0: 800x800 (no detections), 12.1ms
Speed: 5.8ms preprocess, 12.1ms inference, 0.5ms
0: 800x800 (no detections), 12.2ms
Speed: 4.8ms preprocess, 12.2ms inference, 0.7ms
Basketball Hoop Detection (on 0694fdbd00f0)
FPS: 29.53
(x=241, y=4) ~ R:15 G:16 B:12